

This condensed reference summarizes the essential equations and stability function formulations from England & McNider (1995), correcting the quadratic root sign error for continuous stability transitions.

1. Exchange Coefficients & Surface Layer Framework

- **Monin–Obukhov Exchange Coefficients:**

$$K_m = \frac{l_m u_*}{\phi_m} = \frac{l_m^2 s}{\phi_m^2}, \quad K_h = \frac{l_m u_*}{\phi_h} = \frac{l_m^2 s}{\phi_m \phi_h}$$

- **Friction Velocity & Gradients:**

$$u_* = \frac{l_m s}{\phi_m}, \quad s = \left[\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2 \right]^{1/2}, \quad l_m = \kappa z, \quad \theta_* = \frac{l_m}{\phi_h} \frac{\partial \theta}{\partial z}$$

- **Dimensionless Shear Functions $\phi(\zeta)$:**
- *Stable* ($\zeta > 0$): $\phi_m(\zeta) = 1 + \beta_m \zeta$, $\phi_h(\zeta) = 1 + \beta_h \zeta$
- *Unstable* ($\zeta < 0$): $\phi_m(\zeta) = (1 - \beta_m \zeta)^{-1/4}$, $\phi_h(\zeta) = (1 - \beta_h \zeta)^{-1/2}$

2. Gradient Richardson Number (Ri) & Inverse Relations

- **Gradient Richardson Number:**

$$Ri = \frac{g}{\theta_0} \frac{\partial \theta / \partial z}{s^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)}$$

- **Stable Branch Quadratic Equation:**

$$(Ri \beta_m^2 - \beta_h) \zeta^2 + (2Ri \beta_m - 1) \zeta + Ri = 0$$

- **Continuous Physical Solution $\zeta(Ri)$ (Corrected Sign):**

$$\zeta = \frac{1 - 2Ri \beta_m - \sqrt{1 + 4(\beta_h - \beta_m) Ri}}{2(Ri \beta_m^2 - \beta_h)}$$

- **Critical Richardson Number & Simplified Form ($\beta_h = \beta_m$):**

$$Ri_c = \frac{\beta_h}{\beta_m^2}, \quad \zeta = \frac{Ri}{1 - Ri/Ri_c}$$

- **Unstable Branch Cubic Equation:**

$$\beta_m \zeta^3 - \zeta^2 - \beta_h Ri^2 \zeta + Ri^2 = 0$$

3. Stability Functions $f_m(Ri)$ & $f_h(Ri)$

- **Exchange Coefficients in Terms of Ri :**

$$K_m = f_m(Ri) l_m^2 s, \quad K_h = f_h(Ri) l_m^2 s$$

$$f_m(Ri) = \phi_m^{-2}(\zeta), \quad f_h(Ri) = \phi_m^{-1}(\zeta) \phi_h^{-1}(\zeta)$$

- **Stable Layer Quadratic Form ($Pr = 1, \beta_m = \beta_h$):**

$$f_m(Ri) = f_h(Ri) = \begin{cases} \left(1 - \frac{Ri}{Ri_c}\right)^2, & 0 \leq Ri \leq Ri_c \\ 0, & Ri > Ri_c \end{cases}$$

- **Blackadar Linear Form (Comparison):**

$$f_m(Ri) = f_h(Ri) = \begin{cases} 1 - \frac{Ri}{Ri_c}, & 0 \leq Ri \leq Ri_c \\ 0, & Ri > Ri_c \end{cases}$$

Correction Note

Equation (11) in the historical draft contained a plus sign before the square root. Selecting the minus sign in $\zeta = \frac{1-2Ri\beta_m - \sqrt{1+4(\beta_h-\beta_m)Ri}}{2(Ri\beta_m^2-\beta_h)}$ is mathematically required to guarantee $\zeta \rightarrow 0$ as $Ri \rightarrow 0$ at neutral stability.